1. Log in to Cisco Packet Tracer or any network simulation tool, add a switch, and create two VLANs named 'Sales' and 'Support' with VLAN IDs 10 and 20.

ANS:-

Creating VLANs in Cisco Packet Tracer

Aim: To create two VLANs named Sales and Support on a switch.

Commands:

Switch> enable

Switch# configure terminal

Switch(config)# vlan 10

Switch(config-vlan)# name Sales

Switch(config-vlan)# exit

Switch(config)# vlan 20

Switch(config-vlan)# name Support

Switch(config-vlan)# exit

Switch(config)# end

Explanation:

- VLAN 10 is named Sales.
- VLAN 20 is named Support.
- VLANs logically divide a switch network into separate broadcast domains.

Result: Two VLANs, Sales and Support, are created successfully.

2. Assign three different switch ports to VLAN 10 (Sales) and two ports to VLAN 20 (Support), then verify the VLAN membership using the show vlan command. Hint: Use the interface range command to select multiple ports efficiently.

ANS:-

Aim: To assign three switch ports to VLAN 10 and two ports to VLAN 20.

Commands:

Switch# configure terminal

Switch(config)# interface range fa0/1-3

Switch(config-if-range)# switchport mode access

Switch(config-if-range)# switchport access vlan 10

Switch(config-if-range)# exit

Switch(config)# interface range fa0/4-5

Switch(config-if-range)# switchport mode access

Switch(config-if-range)# switchport access vlan 20

Switch(config-if-range)# exit

Switch(config)# end

Switch# show vlan brief

Expected VLAN membership:

VLAN ID	VLAN name	Assigned ports
10	Sales	Fa0/1, Fa0/2, Fa0/3
20	Support	Fa0/4, Fa0/5

Explanation: The interface range command selects multiple switch ports at once. The switchport access vlan command assigns the selected ports to the required VLAN.

Result: The show vlan brief command verifies the VLAN names, IDs, and assigned access ports.

3. Connect two routers using a serial or Ethernet link in your simulation tool, assign IP addresses to their interfaces, and configure a static route on Router 1 to reach the LAN behind Router 2.

ANS:-

Static Routing Between Two Routers:

Router 1 Configuration:

Router1> enable

Router1# configure terminal

Router1(config)# interface gigabitEthernet0/0

Router1(config-if)# ip address 192.168.1.1 255.255.255.0

Router1(config-if)# no shutdown

Router1(config-if)# exit

Router1(config)# interface gigabitEthernet0/1

Router1(config-if)# ip address 10.0.0.1 255.255.255.252

Router1(config-if)# no shutdown

Router1(config-if)# exit

Router 2 Configuration:

Router2> enable

Router2# configure terminal

Router2(config)# interface gigabitEthernet0/0

Router2(config-if)# ip address 10.0.0.2 255.255.255.252

Router2(config-if)# no shutdown

Router2(config-if)# exit

Router2(config)# interface gigabitEthernet0/1

Router2(config-if)# ip address 192.168.2.1 255.255.255.0

Router2(config-if)# no shutdown

Router2(config-if)# exit

Static Route on Router 1

The LAN behind Router 2 is 192.168.2.0/24, so Router 1 is configured with:

Router1(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2

Result: Router 1 can now reach the 192.168.2.0/24 LAN through Router 2 at 10.0.0.2.

4. Simulate a scenario where a device in the 'Sales' VLAN tries to ping a device in the 'Support' VLAN and explain why the ping fails, then suggest what configuration would allow inter-VLAN communication. Hint: Think about router-on-a-stick or Layer 3 switch concepts.

ANS:-

Testing Communication Between Sales and Support VLANs

Scenario: A PC in VLAN 10 (Sales) attempts to ping a PC in VLAN 20 (Support).

Why does the ping fail?

VLAN 10 and VLAN 20 are separate Layer 2 broadcast domains. A normal Layer 2 switch does not route traffic between them, so devices in different VLANs cannot communicate directly without a Layer 3 routing configuration.

Solution: Enable inter-VLAN routing.

Two common methods are:

1. Router-on-a-stick: Configure a trunk link between the switch and a router, create router subinterfaces for VLANs 10 and 20, and assign each subinterface an IP address to act as its VLAN's default gateway.
2. Layer 3 switch: Configure switched virtual interfaces (SVIs) for VLANs 10 and 20, assign gateway IP addresses, and enable IP routing.

Result: After inter-VLAN routing and correct IP addressing and default gateways are configured, devices in the Sales and Support VLANs can communicate, subject to any access-control rules.